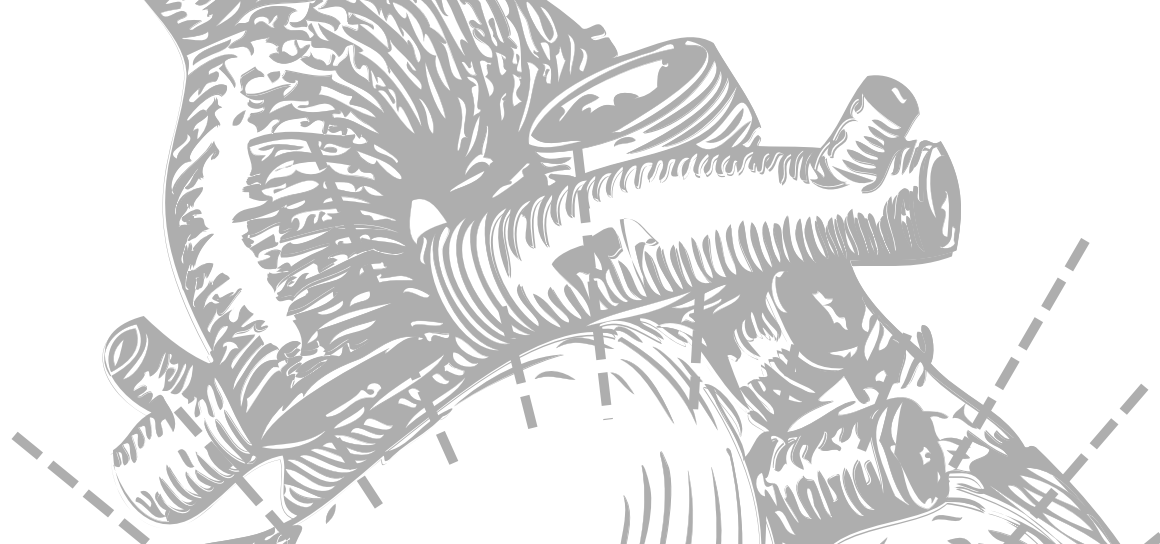
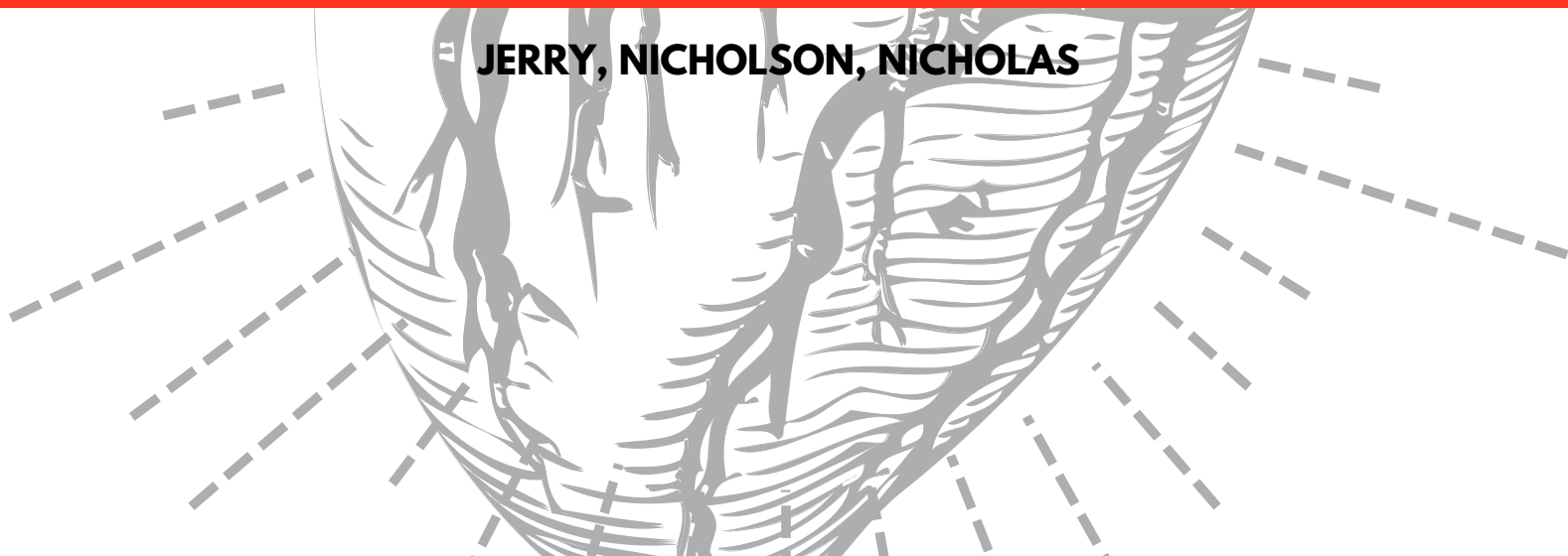




PREDICTING CVD RISK USING SLEEP AND HEALTH DATA

JERRY, NICHOLSON, NICHOLAS





Agenda

1. Introduction

2. Dataset Overview

**3. Feature Selection &
Criteria**

4. Model Operations

5. Shap Analysis

**6. Findings &
Deployment**

WHAT IS CVD ?

Cardiovascular disease (CVD) refers to conditions involving the heart and blood vessels — including heart attacks, strokes, hypertension-related diseases, and heart failure. It is the #1 cause of death globally — 17.9 million lives lost each year.



GAP !

- Most existing clinical tools do not **integrate** sleep data with health indicators.
- Sleep disorders are common but **underdiagnosed**
- Data exist but is **underutilized clinically**

RESEARCH QUESTIONS



PRIMARY:

DO SLEEP AND HEALTH DATA PREDICT CVD RISK?



SECONDARY 1:

WHICH SLEEP AND HEALTH FACTORS DRIVE CVD MOST ?

- DO SLEEP METRICS ADDED PREDICTIVE VALUE ?

SECONDARY 2:

WHICH VARIABLES ARE THE MOST INFLUENTIAL
PREDICTORS OF CVD RISK BASED ON MODEL ?



DATASET OVERVIEW

DATA PREP

Sleep Heart Health Study (SHHS)

Datasets:

- Overnight sleep study data
 - 1995 - 1998
 - (5804, 1279)
- 16y CVD disease data
 - 1995 - 2011
 - (5802, 41)

---- inner joint on shared **Patient ID** ----

First 5 rows:

nsrrid	age_s1	bmi_s1	SystBP	any_cvd
200001	55	21.777553	143.0	NaN
200002	78	32.950680	168.0	NaN
200003	77	24.114150	127.0	NaN
200004	48	20.185185	130.0	NaN
200005	66	23.309053	145.0	NaN

Result: (5802, 1280)

DATA CLEANING

- target : any_cvd
 - (0 = No CVD, 1 = CVD)
- missing any_cvd dropped
 - now 5042 → 760 rows
- numeric imputation → median
- IQR Crapping to 11 columns
- rename 19 columns
- no duplicate rows found



FEATURE SELECTION

1,280 VARIABLES



REMOVE LEAKAGE



DOMAIN KNOWLEDGE



18 VARIABLES

**Dropped ID columns, date columns,
and all CVD outcome columns**

**Reviewed SHHS data dictionary and
existing CVD literature**



FINAL DATASET

DATA PIPELINE → FINAL DATASET

- 1 Load & Merge**
Inner join on nsrrid → 5,802 rows × 1,280 cols
- 2 Feature Selection**
Select 18 clinically relevant features + target (any_cvd)
- 3 Data Cleaning**
Drop 760 rows (missing target) · Median imputation · IQR outlier capping
- 4 Rename Features**
Short SHHS codes → descriptive English names (e.g. ahi_a0h3 → apnea_events_per_hour)
- 5 EDA**
T-test significance analysis · CVD rate by sex · Correlation heatmap visualization
- 6 Final Dataset**
5,042 rows × 19 cols · CVD=1: 23.7% · CVD=0: 76.3%

Final Dataset Summary

Rows:	5,042
Columns:	19 (18 features + target)
Sleep features:	5
Health features:	13
CVD=1 (positive):	1,196 (23.7%)
CVD=0 (negative):	3,846 (76.3%)

PERFORMANCE CRITERIA

**AUC
> 0.75**



**Matches the best published SHHS results.
Below 0.75 = not better than guessing.**

**Recall
> 0.70**



**7 out of 10 CVD patients caught.
Miss a patient = no treatment.**

**Accuracy
> 76%**



**Beat 76% Accuracy Predicting "No CVD"
every time already hits 76%.
Our model must do better.**



MODEL OPS



3 MODEL SELECTIONS



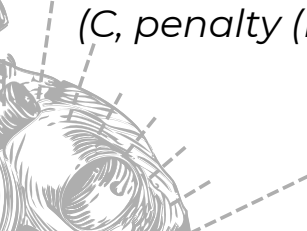
Logistic Regression

- **Strong** baseline model for comparisons
- **Highly interpretable** - easy to explain to doctor/clinical
- Test whether **relationships** are linear and predictable



Interpretability

(C, penalty (L1/L2), solver)



Random Forest

- Captures **non-linear** relationships automatically
- Handles feature **interactions** (e.g., combined risk factors)
- More **robust** and **less prone** to overfitting



Complex Patterns

(n_estimators, max_depth, min_samples_split)

Gradient Boosting

- Builds models **sequentially** to improve errors
- Typically delivers **high predictive performance**
- Proven strong for **structured/tabular data**



for Best Performance

(n_estimators, learning_rate, max_depth, subsample)

MODEL ASSUMPTION VALIDATION

```
--- VIF Results (numeric features, sorted descending) ---  
      Feature  VIF  
sleep_efficiency_pct 4.50  
wake_after_sleep_onset_min 4.47  
      bmi 2.65  
waist_circumference_cm 2.63  
min_oxygen_saturation 2.36  
apnea_events_per_hour 1.85  
avg_oxygen_saturation 1.82  
      diastolic_bp 1.74  
age_at_baseline 1.73  
      systolic_bp 1.66  
hdl_cholesterol 1.18  
      cholesterol 1.04  
  
No VIF > 10 – multicollinearity is acceptable for all features
```

```
--- Data Leakage Check ---  
No CVD outcome columns in feature set – data leakage confirmed absent
```

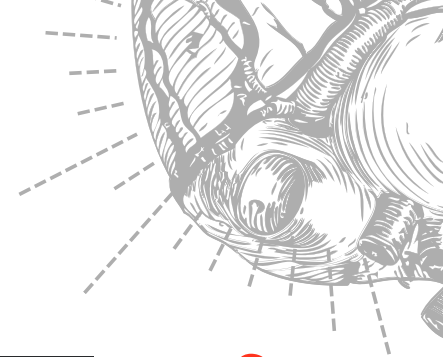
```
--- Training Set Class Balance (after undersample) ---  
any_cvd=0: 957 (50.0%)  
any_cvd=1: 957 (50.0%)  
Ratio: 1.00 (target: 1.00 = perfectly balanced)
```

VIF Check Max VIF = 4.50.
All features well below threshold of 10. No multicollinearity.

Data Leakage Check Zero CVD outcome columns in the feature set. No leakage detected

Class Balance 957 CVD / 957 No CVD after undersampling.
Perfectly balanced for training.

HYPERPARAMETER TUNING



**Before
Tuning**

Model	K=3 Mean	K=3 Std	K=5 Mean	K=5 Std	K=10 Mean	K=10 Std
Logistic Regression	0.7911	0.0068	0.7894	0.0195	0.7888	0.0248
Random Forest	0.7663	0.0132	0.7651	0.0244	0.7718	0.0314
Gradient Boosting	0.7681	0.0116	0.7684	0.0127	0.7755	0.0254

After K selection, each model was tuned with RandomizedSearchCV
Best params selected by **highest roc_auc**.

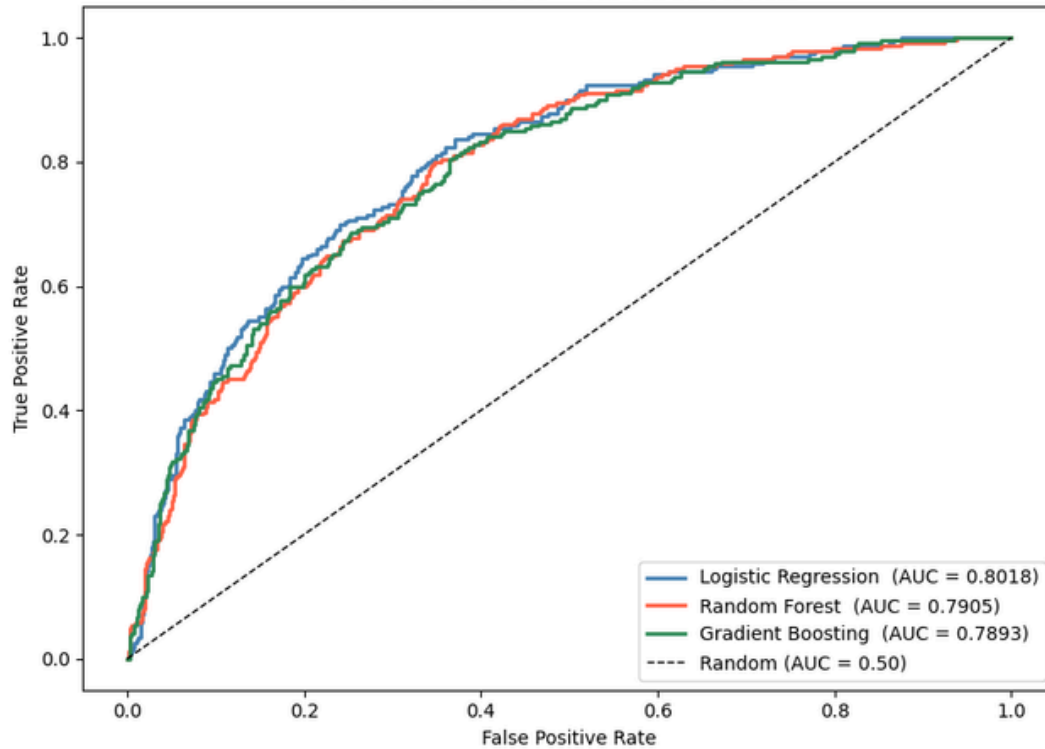
**After
Tuning**

Model	CV AUC	Test AUC	Recall	F1	Threshold	Time
Logistic Regression	0.7914	0.8018	0.7071	0.5624	0.55	0.8s
Random Forest	0.7838	0.7905	0.7071	0.5391	0.51	13s
Gradient Boosting	0.7820	0.7893	0.7113	0.5304	0.51	82s

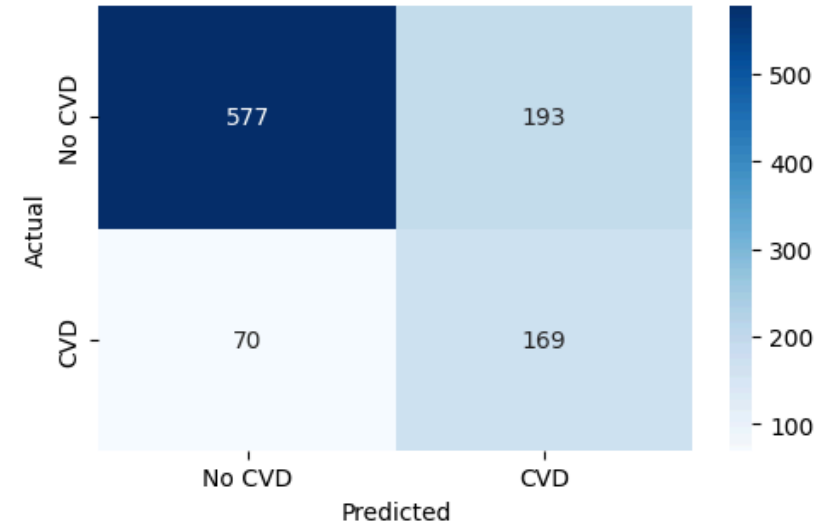


MODEL COMPARISON

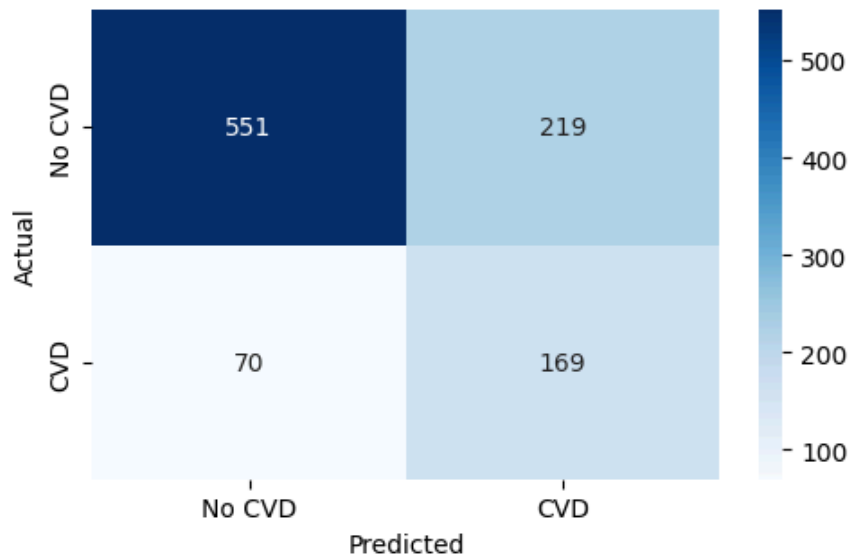
AUC-ROC Curves — All Three Models



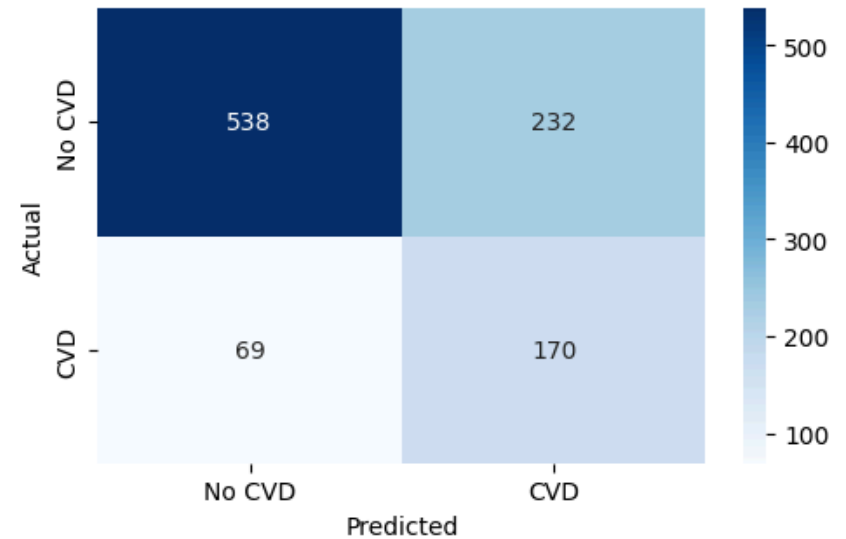
Logistic Regression
(threshold=0.55)



Random Forest
(threshold=0.51)



Gradient Boosting
(threshold=0.51)



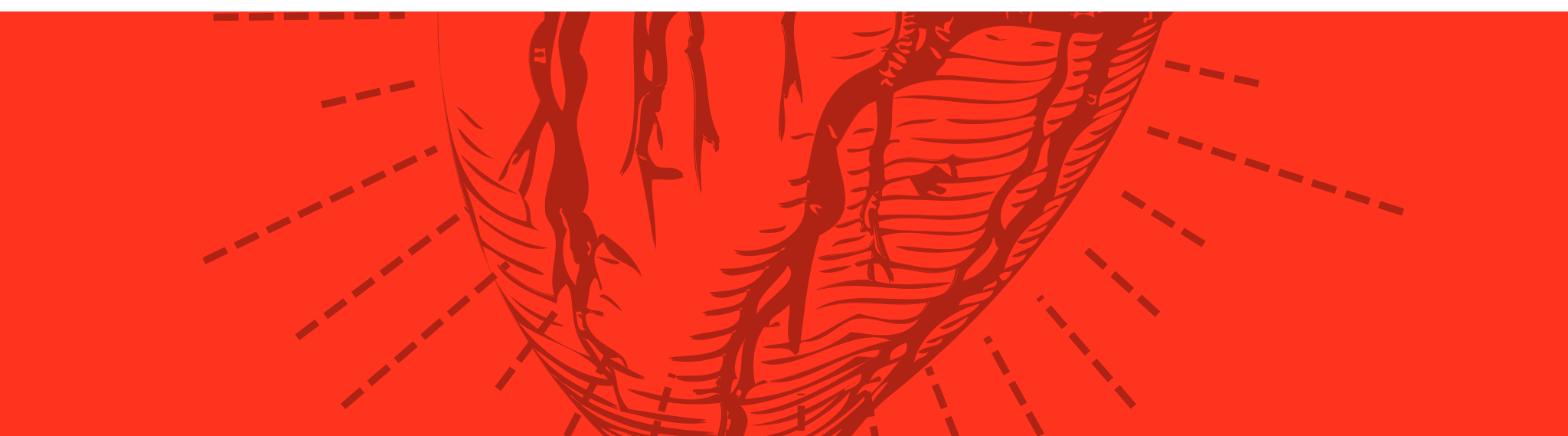
FINAL MODEL

Model	CV AUC	Test AUC	Recall	F1	AUC $\geq$ 0.75	Recall $\geq$ 0.70	Train Time
Logistic Regression	0.7914	0.8018 ★	0.7071	0.5624 ★	PASS	PASS	0.8s ★
Random Forest	0.7838	0.7905	0.7071	0.5391	PASS	PASS	13s
Gradient Boosting	0.782	0.7893	0.7113 ★	0.5304	PASS	PASS	82s

1. Highest AUC (**0.8018**) and F1 (**0.5624**)
2. Most interpretable: easy to explain clinically
3. All 3 models pass: LR wins every secondary metric (F1 score, Train Time)
4. GB Recall diff = 0.004, not worth 100x train time



SHAP ANALYSIS



SHAP ANALYSIS

Top 5 Sleep Metrix

Apnea events/hr	0.0886
Min O2 saturation	0.0309
Sleep efficiency	0.0189
Avg O2 saturation	0
Wake after sleep	0



Top 5 Health Indicator

Age	0.7336
Sex	0.4002
BP Medication	0.3408
Smoking Status	0.1689
HDL Cholesterol	0.1531



Rank	Type	Variable	SHAP
1	Health	Age at baseline	0.7336
2	Health	Sex	0.4002
3	Health	Taking BP medication	0.3408
4	Health	Smoking status	0.1689
5	Health	HDL cholesterol	0.1531
6	Health	Systolic BP	0.1463
7	Health	Diastolic BP	0.1385
8	Sleep	Apnea events per hour	0.0886
9	Health	Waist circumference	0.0669
10	Health	History of diabetes	0.0597
11	Health	Self-reported hypertension	0.0521
12	Health	Race	0.0428
13	Sleep	Min oxygen saturation	0.0309
14	Health	Cholesterol	0.0202
15	Health	BMI	0.02
16	Sleep	Sleep efficiency	0.0189
17	Sleep	Avg oxygen saturation	0
18	Sleep	Wake after sleep onset	0



KEY VARIABLES & THEIR IMPACT ON CVD RISK



HEALTH FACTORS (MAIN DRIVERS)

Variables		Impact on CVD Risk	
	Age at baseline	→	Risk increases with age.
	Sex	→	Males tend to have higher early risk.
	Taking BP medication	→	Indicates existing hypertension.
	Smoking status	→	Damages blood vessels and increases risk.
	HDL cholesterol	→	Low levels of “good” cholesterol increase risk.
	Systolic BP	→	High top-number pressure strains the heart.
	Diastolic BP	→	High bottom-number pressure increases risk.
	Waist circumference	→	Linked to abdominal obesity and metabolic risk.

 **Note:** Age at baseline, Sex, and Taking BP medication are part of the “Variables”.



SLEEP FACTOR (ADDITIONAL INSIGHT)



Apnea events per hour



Frequent breathing interruptions reduce oxygen levels during sleep.



Increases stress on the heart and cardiovascular system.



KEY TAKEAWAY:
Health factors are the **dominant drivers** of CVD risk, while **sleep (especially apnea)** provides additional but smaller impact.

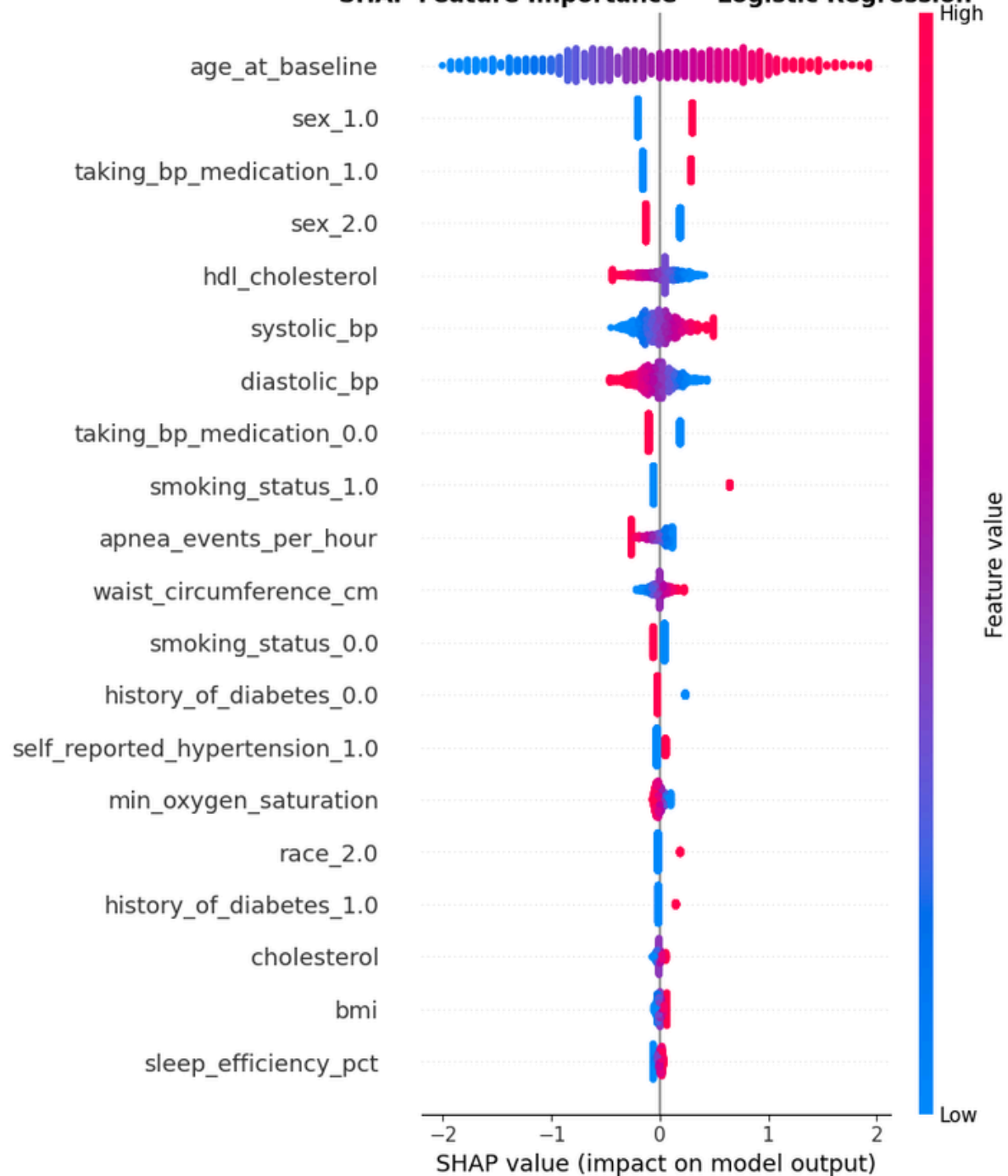




FINDINGS

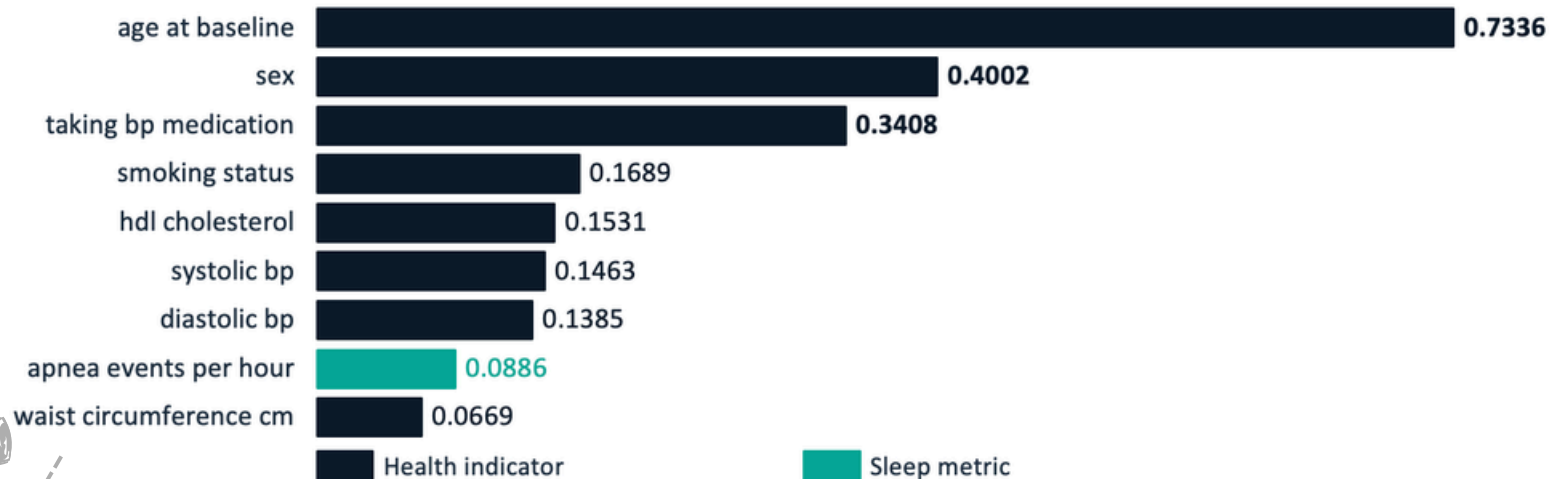


SHAP Feature Importance — Logistic Regression



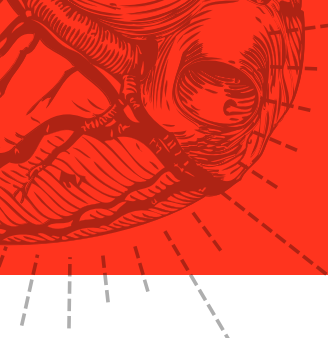
KEY FEATURE DRIVERS

SHAP Feature Importance (Mean |SHAP|)



WHY SELECTED FEATURES?

- Highest SHAP importance
- Removes noise from low-impact variables
- Improves interpretability
- Reduces overfitting
- Maintains performance
- Top features capture majority of predictive power
- Health factors dominate; sleep variables have smaller but meaningful impact



CONCLUSIONS

KEY FINDINGS

- Health > Sleep in prediction
- Strong performance (AUC 0.80)
- Age = top driver
- Apnea = top sleep factor

LIMITATIONS

- Data is outdated (1995–1998)
- Lower performance for patients under 65
- Designed for prediction, not clinical decisions

FUTURE WORKS

- Validate using more recent data (post-2010)
- Explore advanced models to improve performance
- Add more detailed health history features

SUBGROUP ANALYSIS					
Subgroup	n	AUC	Recall	AUC Drop	Flag
Age < 65	513	0.7289	0.2449	9.1%	YES
Age >= 65	496	0.7208	0.8263	10.1%	
Sex=1	448	0.7579	0.7903	5.5%	
Sex=2	561	0.8422	0.6174	-5.0%	

Model correctly identifies
7 out of 10 CVD patients



ANSWERING THE RESEARCH QUESTIONS

Q1: DO SLEEP AND HEALTH DATA PREDICT CVD RISK?

ANSWER : YES — AUC 0.8018 CONFIRMS STRONG DISCRIMINATION. THE MODEL IDENTIFIES 7 OUT OF 10 CVD PATIENTS.

Q2 (SECONDARY 1): WHICH SLEEP AND HEALTH FACTORS DRIVE CVD MOST?

ANSWER : AGE, SEX, AND BLOOD PRESSURE DOMINATE. APNEA EVENTS/HOUR IS THE STRONGEST SLEEP PREDICTOR BUT RANKS 9TH OVERALL.

Q3 (SECONDARY 2): WHICH VARIABLES ARE THE MOST INFLUENTIAL OF CVD RISK BASED ON MODEL?

ANSWER : TOP 9 VARIABLES OVERALL — AGE, SEX, BP MEDICATION, HYPERTENSION, SYSTOLIC BP, DIASTOLIC BP, CHOLESTEROL, HDL, APNEA EVENTS/HOUR.

SUBGROUP FINDING: MODEL PERFORMS WELL FOR AGE 65+ (RECALL 82.6%) AND FEMALES (AUC 0.842). RECALL DROPS TO 24.5% FOR PATIENTS UNDER 65 — A KNOWN LIMITATION.



THANK YOU

EDA-HEAT MAP



EDA - T-TEST AND SEX ANALYSIS

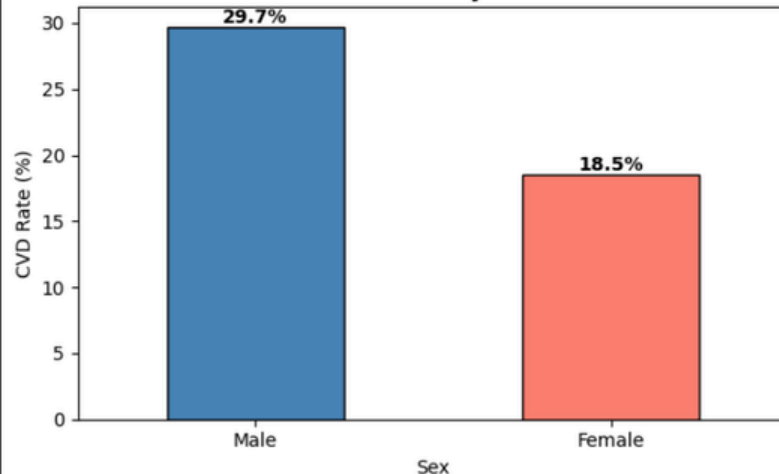
... === T-test Results: CVD=1 vs CVD=0 ===

Feature	Mean CVD	Mean No CVD	p-value	Result
apnea_events_per_hour	16.551	12.761	0.0000	significant
avg_oxygen_saturation	93.976	94.564	0.0000	significant
min_oxygen_saturation	85.079	86.098	0.0000	significant
sleep_efficiency_pct	80.454	83.924	0.0000	significant
wake_after_sleep_onset_min	71.600	57.578	0.0000	significant
bmi	28.261	28.113	0.3491	not significant
waist_circumference_cm	99.703	96.584	0.0000	significant
systolic_bp	130.839	123.510	0.0000	significant
diastolic_bp	69.945	72.797	0.0000	significant
cholesterol	206.225	206.108	0.9212	not significant
hdl_cholesterol	48.229	50.634	0.0000	significant
age_at_baseline	71.247	61.602	0.0000	significant

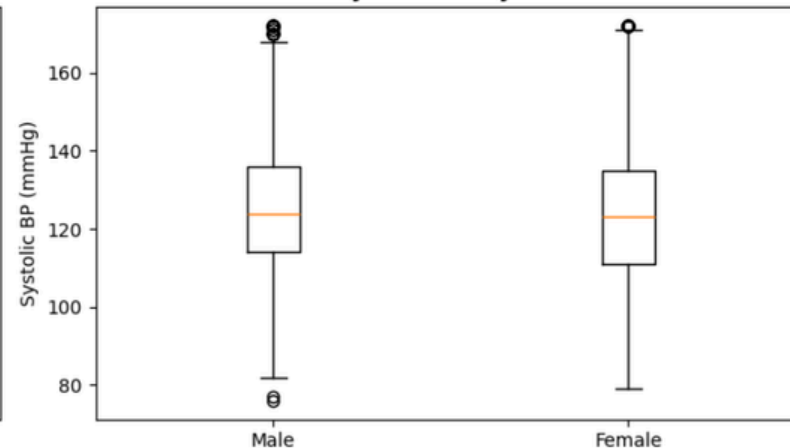
... === CVD Rate by Sex ===

	CVD Count	Total Patients	CVD Rate (%)
sex			
Male	698.0	2349	29.7
Female	498.0	2693	18.5

CVD Rate by Sex



Systolic BP by Sex



DEFAULT THRESHOLD - EACH MODEL

Scanned thresholds from 0.90 down to 0.05 and picked the highest value that still kept Recall at or above 0.70. That's how we got 0.55 for LR and 0.51 for RF and GB.

```
***
Logistic Regression (threshold=0.55)
      precision    recall  f1-score   support

   No CVD       0.89      0.75      0.81       770
    CVD        0.47      0.71      0.56       239

 accuracy          0.74      1009
 macro avg       0.68      0.73      0.69      1009
weighted avg       0.79      0.74      0.75      1009

AUC: 0.8018

Random Forest (threshold=0.51)
      precision    recall  f1-score   support

   No CVD       0.89      0.72      0.79       770
    CVD        0.44      0.71      0.54       239

 accuracy          0.71      1009
 macro avg       0.66      0.71      0.67      1009
weighted avg       0.78      0.71      0.73      1009

AUC: 0.7905

Gradient Boosting (threshold=0.51)
      precision    recall  f1-score   support

   No CVD       0.89      0.70      0.78       770
    CVD        0.42      0.71      0.53       239

 accuracy          0.70      1009
 macro avg       0.65      0.70      0.66      1009
weighted avg       0.78      0.70      0.72      1009

AUC: 0.7893
```

CONCLUSIONS

KEY FINDINGS

- Sleep metrics only account for 5.6% of total SHAP importance, while Health indicators account for 94.4%.
- AUC 0.8018 matches the best published SHHS research.
- Age is the single strongest predictor; apnea events are the top sleep predictor

Model correctly identifies
7 out of 10 CVD patients

LIMITATIONS

- Data is limited (1995 - 1998)
- Underperform / ineffective for patients under 65 (recall = 24.5%)
- Model is designed for predictive analysis, not clinical decision-making

SUBGROUP ANALYSIS

Subgroup	n	AUC	Recall	AUC Drop	Flag
Age < 65	513	0.7289	0.2449	9.1%	
Age >= 65	496	0.7208	0.8263	10.1%	YES
Sex=1	448	0.7579	0.7903	5.5%	
Sex=2	561	0.8422	0.6174	-5.0%	

FUTURE WORKS

- Validate on more recent datasets (post-2010) to test more generalizability.
- Explore LightGBM and Catboost for potential AUC improvements
- Add medication history and longitudinal health records to enrich the feature set.